

1

Main Use Case Descriptions

USE CASE DESCRIPTION - 1

Add Shared Expense

This use case allows a roommate user to add a shared household expense within a roommate group. The user enters expense details such as amount, category, participants, and split type. The system validates the input, calculates each roommate's share, updates balances, and stores the expense information for future tracking and settlements.

USE CASE DESCRIPTION - 2

Manage Chores

This use case allows a roommate user to create, assign, and track household chores among roommates. Users can assign tasks with due dates and recurrence settings, while the system updates chore status and sends reminders to assigned roommates to maintain fair responsibility distribution.

USE CASE DESCRIPTION - 3

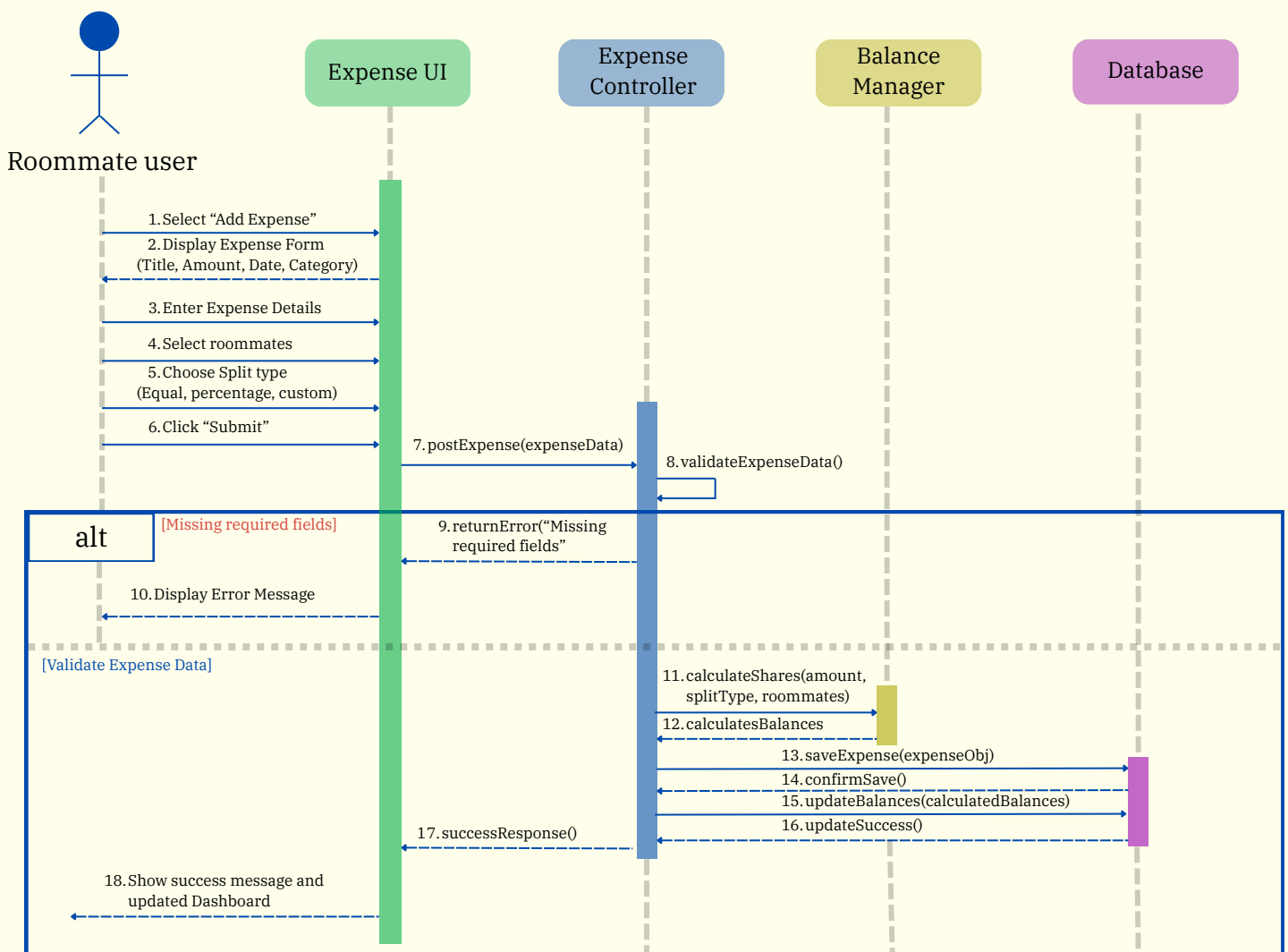
Report Maintenance Issue

This use case allows a roommate user to report maintenance-related problems in the shared home environment. Users can provide issue details, priority level, and optional images. The system records the issue, tracks its status, and notifies roommates about updates until the issue is resolved.

2 Sequence Diagram

SEQUENCE DIAGRAM

Add Shared Expense



Legend:



Synchronous message



Return Message

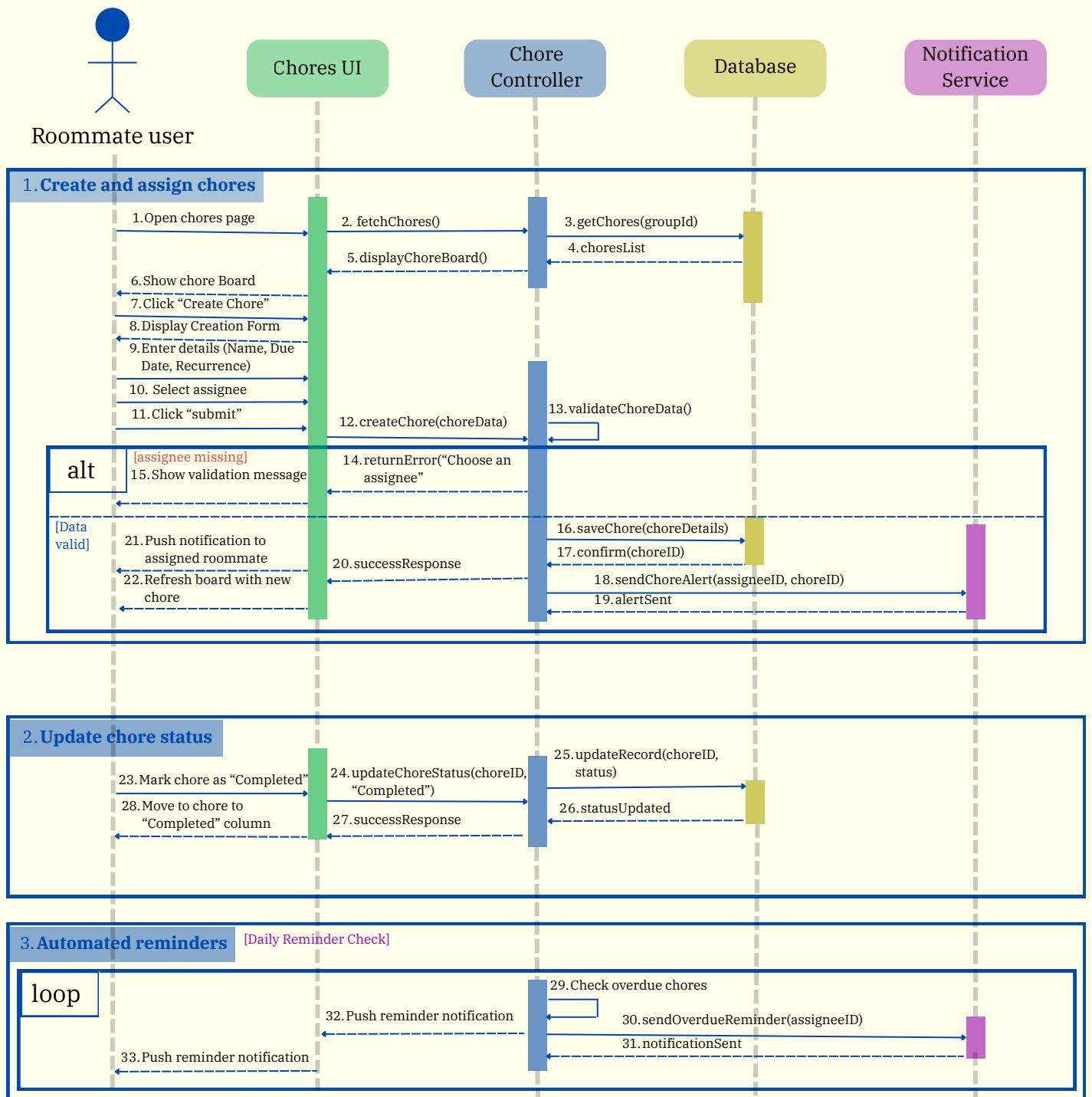
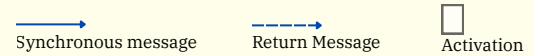


Activation

2 Sequence Diagram

SEQUENCE DIAGRAM Manage Chores

Legend:



2 Sequence Diagram

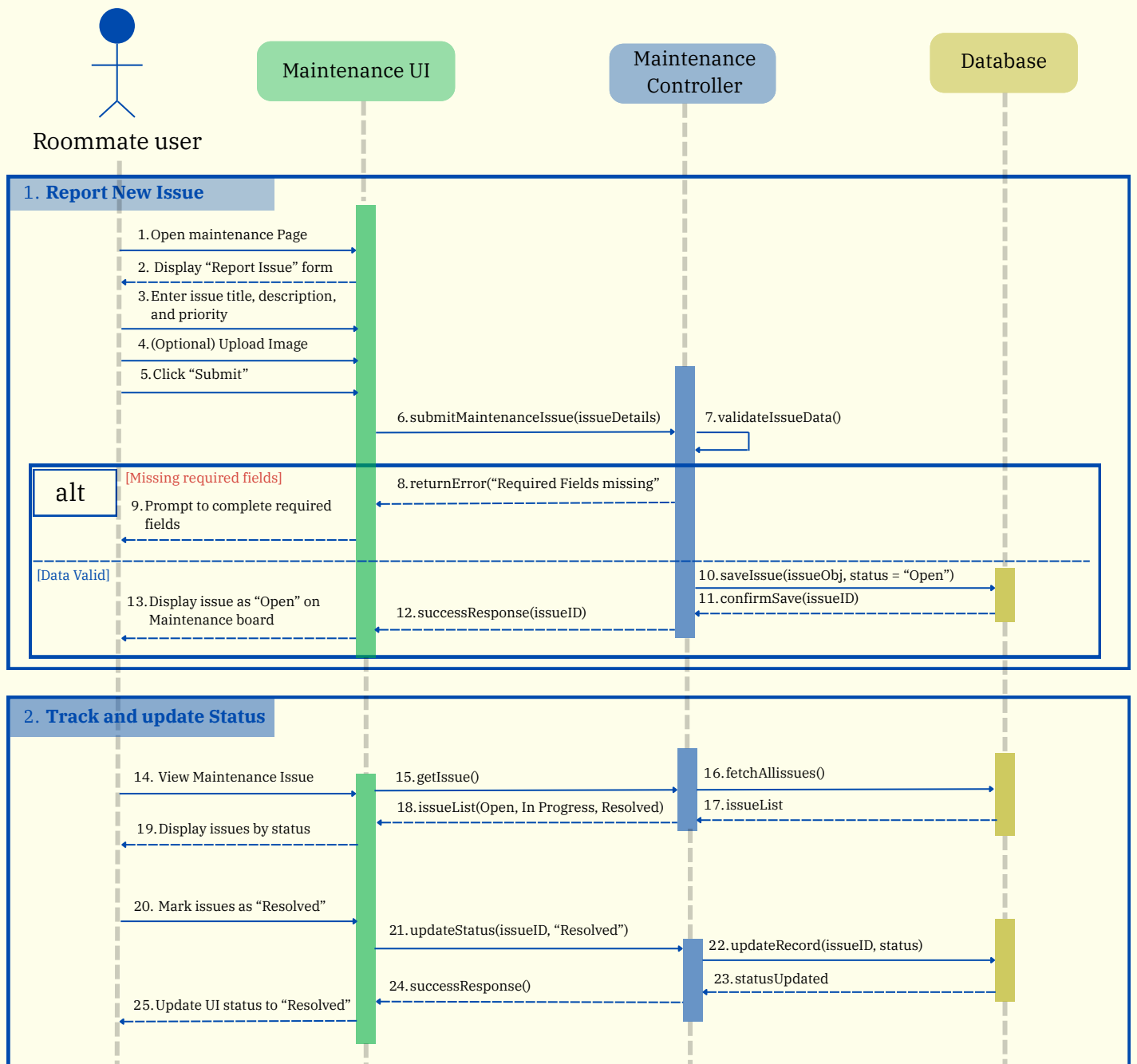
SEQUENCE DIAGRAM Report Maintenance Issue

Legend:

—→ Synchronous message

- - -> Return Message

□ Activation



3 Class Diagram

